

Pearson Edexcel International GCSE ICT (4IT1)

Paper 1: Digital Communication Mock Examination

prepared by Tr Jamal Fahadh for ICT candidates, Best academy, Hargeisa Somaliland

Total Marks: 150 Marks

Question 1 (Total: 30 marks)

A multimedia production studio relies on various data transmission technologies to manage network files and broadcast live feeds.

(a) State what is meant by **bandwidth**. (1)

Bandwidth definition...

(b) Define the term **latency**. (1)

Latency definition...

(c) Network engineers use a specific technique to measure connection latency. Identify this measurement technique and state how it determines the delay time. (2)

Measurement technique and mechanism...

(d) A user experiences visible "lag" while playing a real-time multiplayer online game. Explain how bandwidth and latency individual configurations alter the user experience during online gaming or HD video streaming. (4)

Impact of Bandwidth:

Impact of Latency:

(e) The studio streams high-definition video over its network.

(i) State the function of a **buffer** during digital video playback. (1)

Function of buffer...

(ii) Describe the underlying mechanism that background downloads must maintain to prevent pausing lag. (2)

Underlying mechanism...

(f) Explain three environmental or structural factors that can disrupt wireless communication performance inside an office building. (6)

Factor 1:

Factor 2:

Factor 3:

(g) Cabled connection wires use thin metal layers to preserve signal data integrity.

(i) State the technical term for these protective layers. (1)

Technical term...

(ii) Describe the primary function of this structural design element. (2)

Primary function...

(h) Global video game companies deploy dedicated gaming servers across multiple geographic regions. Explain why utilizing servers in multiple countries is necessary to ensure the user experience is not negatively affected. (3)

Explanation...

(i) Identify one statement that best describes the ultimate wide area network (WAN). Place a cross [X] in the correct box. (1)

- ☐ A. An office building intranet
- ☐ B. A short-range personal device connection
- ☐ C. The Internet
- ☐ D. A localized broadband bridge

(j) Contrast how a video streaming configuration handles data files compared to traditional downloading methods from a user perspective. (6)

Streaming:

Downloading:

Question 2 (Total: 30 marks)

Digital systems are categorized into distinct structural network frameworks based on their geographic spans and hardware roles.

(a) Complete the table below by detailing the parameters and features of the four major network classifications. (8)

Network Type	Geographical Range	Primary Characteristics & Contexts
LAN	Small, restricted space	(i)
WAN	(ii)	Links disparate corporate or government networks together.
PAN	Immediately surrounding a person	(iii)
Tethering	Short-range localized bridge	(iv)

(b) An employee wants to share their smartphone's mobile data connection with a laptop while working on a train. Describe how the employee can configure the smartphone to serve this function. (2)

Configuration steps...

(c) Wired topologies rely on specific connection interfaces to manage peripheral data buses. Identify the correct primary use case from the syllabus guidelines for the following wired media standards: (5)

• **HDMI:**

Primary use case for HDMI...

• **S/PDIF:**

Primary use case for S/PDIF...

• **Minijack (3.5 mm):**

Primary use case for Minijack (3.5 mm)...

• **USB:**

Primary use case for USB...

• **Ethernet:**

Primary use case for Ethernet...

(d) A computer lab technician chooses to connect all workstation nodes using Ethernet cables rather than Wi-Fi routers. Explain three distinct performance or operational advantages of choosing wired configurations over wireless bands. (6)

Advantage 1:

Advantage 2:

Advantage 3:

(e) Despite performance advantages, physical wires introduce specific operational constraints inside modern workspaces. State two distinct drawbacks associated with installing complex cabled infrastructure. (2)

1.

2.

(f) Identify one core protocol standard specification that dictates the architectural rules of Wi-Fi networks. Place a cross [X] in the correct box. (1)

- ☐ A. IEEE 802.11
- ☐ B. DVB-S2
- ☐ C. RFID Proximity
- ☐ D. DAB Text

(g) Differentiate between how copper media paths transmit raw data signals compared to open wireless communication bands. (6)

Detailed comparative analysis...

Question 3 (Total: 30 marks)

Wireless communication infrastructure is deployed across various consumer devices, industrial settings, and global transit networks.

(a) Satellite networks deliver comprehensive coverage for multiple critical tracking systems.

(i) State two benefits of utilizing satellite communication systems for remote digital data transmission. (2)

1.

2.

(ii) State two environmental or structural drawbacks that degrade or completely block satellite signals. (2)

3.

4.

(b) The Global Positioning System (GPS) utilizes a specialized network of orbiting satellites.

(i) State the exact minimum number of satellites active within the overall global GPS network. (1)

Minimum number...

(ii) State the exact number of satellite views a terrestrial GPS receiver requires to calculate its location coordinates. (1)

Required satellite views...

(c) DVB-S technology is deployed to distribute video content to residential consumer homes. State the full meaning of the acronym DVB-S and list two peripheral items a viewer needs to decode the signal. (3)

Full Meaning:

Peripheral Item 1:

Peripheral Item 2:

(d) Broadcast transmission stations distribute audio and visual data streams to home receivers.

(i) Contrast DVB-T (Digital Video Broadcasting - Terrestrial) with DVB-S in terms of transmitter hardware locations. (2)

Comparison...

(ii) Identify the standard antenna configuration required by a viewer to receive DVB-T digital signals if they are upgrading from an older analogue setup. (1)

Antenna type...

(iii) State two advanced functionalities provided by the newer DVB-T2 terrestrial broadcast standard. (2)

1.

2.

(e) Digital Audio Broadcasting (DAB) updates traditional radio media streams. Describe how DAB systems operate and state one form of secondary text data a DAB receiver can display alongside audio. (3)

Operation:

Text Data Example:

(f) Short-range wireless standards allow personal consumer items to sync data blocks within close proximity.

(i) State the hardware requirement constraint that Infra-red (IR) communication demands during operation. (1)

Constraint...

(ii) Explain how this operational constraint can affect the user experience if an IR link is used in a video game console controller. (2)

Impact...

(iii) Explain one reason why game console manufacturers choose Bluetooth over Infra-red links for wireless controllers. (1)

Reason...

(g) Near-Field Communication (NFC) architectures are widely implemented across financial infrastructure. Identify the core proximity electronic component category used within NFC frameworks and list two everyday commercial applications. (3)

Component Category:

Application 1:

Application 2:

(h) Cellular mobile carrier configurations supply alternative network paths. Identify two mobile broadband standards used when external local Wi-Fi hotspots are unavailable. (2)

1.

2.

Question 4: Technical Comparison Matrix (Total: 35 marks)

A communications analyst is compiling a matrix evaluating wireless data transmission standards for a client deployment project. Complete the missing properties within the comparison table below by directly sourcing technical criteria from the textbook guidelines. (35 marks, 5 marks per matrix cell)

Feature / Criteria	WI-FI Wireless Connection Standard	BLUETOOTH Personal Standard
Operational Range Profile	Long	(a)
Available Channel Bandwidth	(b)	Low
Power Consumption Footprint	High	(c)
Inherent Security Staging	(d)	Low
Simultaneous Device Links	(e)	(f)
Core Connection Requirement	Connects to a building access point	(g)

Question 5: Extended Response Essay (Total: 25 marks)

A logistics firm is establishing a fleet of delivery vehicles and regional distribution depots. The operations team needs to deploy a communication framework that tracks delivery vehicles in real-time, transfers transit manifests, and syncs handheld inventory systems at regional depots. The infrastructure engineers are evaluating two potential network connection plans:

- **Option A:** Implementing an entirely wireless configuration combining satellite-based tracking (GPS/Voice), cellular data channels (3G/4G) for over-the-road transit, and localized Wi-Fi access hubs at delivery depots.
- **Option B:** Utilizing fixed wired connection channels (Ethernet/USB data cradles) for all data transfers at vehicle depots, requiring drivers to physically plug in systems at the end of each route, and deploying localized Infra-red or Bluetooth pairing tools for tracking changes.

Evaluate both options. You should discuss the impacts of each option on operational data transfer speeds (bandwidth vs. latency bounds), connection stability across varying environments, data security during transit, and overall deployment mobility for the distribution network teams. (25 marks)

Plan your answer carefully. Marks will be awarded for structured layout, correct usage of ICT terms from the text, balanced evaluation arguments, and an effective final conclusion.

Provide your structural analysis and evaluative essay response here...